PREFACE TO APPENDIX A

In this preface to Appendix A, we include some general guidelines for using the "FGDC Digital Cartographic Standard for Geologic Map Symbolization," as well as some basic information on the newly implemented standard for the scientific confidence and the locational accuracy of geologic features. For more specific information, please refer to the following sections in the accompanying introductory text: "Geologic Mapping Concepts and Definitions" (Section 3); "Scientific Confidence and Locational Accuracy of Geologic Features" (Section 4); "Guidelines for Map Color and Pattern Selection" (Section 5); "Guidelines for Map Labeling" (Section 6); and "Technical Specifications Used in the Preparation of this Standard" (Section 7).

ABOUT THIS VERSION OF THE STANDARD

This now formally approved version of the standard incorporates revisions that reflect reviewers' responses to the "Public Review Draft — Digital Cartographic Standard for Geologic Map Symbolization" (Federal Geographic Data Committee, 2000). We thank all the reviewers for their contributions, and we are pleased that most of the comments and suggestions could be accommodated (see *http://ngmdb.usgs.gov/fgdc_gds/geolsymstd/development.php*). Note, however, that we were able to fully address suggestions for the inclusion of new symbols only if examples of the proposed symbol additions were provided.

We intend this standard to be a "living standard" — that is, we recognize that an existing symbol's usage may need to be modified, or a new symbol or set of symbols created, to more fully express local geologic conditions or to keep pace with evolving geologic mapping concepts and practices. Accordingly, such new or modified symbols, if found to be of wide applicability, will be incorporated into this standard through planned, periodic revisions. You are invited to submit comments and suggestions for updates or other improvements to this standard by email to *mapsymbol@flagmail.wr.usgs.gov*.

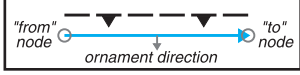
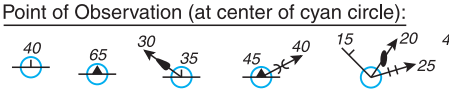
In response to reviewers' comments, we tested various cartographic specifications for dashed and ornamented line symbols in several different software packages (Adobe Illustrator v.8.0.1, ArcInfo v.7x, and ArcGIS v.8x) to ensure that symbols would render correctly and consistently. As a result, we found it necessary to modify the cartographic specifications (dash/gap lengths and ornament spacings) of many line symbols in this standard from the specifications found in previous versions of the standard.

HOW TO USE THIS STANDARD

The contents of this standard are not intended to be used inflexibly or in a manner that will limit one's ability to communicate the observations and interpretations gained from geologic mapping. On the contrary, we recognize that, in certain situations, a symbol or its usage might need to be modified in order to better represent a particular feature on a geologic map or cross section.

To that end, we emphasize that this standard allows the use of any symbol that doesn't conflict with others in the standard, provided that it is clearly explained on the map and in the database. In addition, modifying the size, color, and (or) lineweight of an existing symbol to suit the needs of a particular map or output device also is permitted, provided that the modified symbol's appearance is not too similar to another symbol on the map. Be aware, however, that reducing lineweights below .125 mm (.005 inch) may cause symbols to plot incorrectly if output at higher resolutions (1800 dpi or higher). For more information, please refer to Section 7 ("Technical Specifications Used in the Preparation of this Standard") in the accompanying introductory text.

To facilitate the use of this standard, we include in this preface tables showing conversion values from inches to points to millimeters (Table 1), abbreviations used (Table 2), and spot color specifications and their equivalent colors in other color models (Table 3). We also offer the following illustrations of a few key terms and concepts used when preparing this standard:

LINE SYMBOLS	POINT SYMBOLS	TYPE EXAMPLES	
Right-Hand Rule:	Point of Observation (at center of cyan circle):	Sans-Serif Font:	Serif Font:
		Tv QIs HAYWARD FAULT	Triassic volcanic rocks *Columbia River*

THE NEWLY IMPLEMENTED STANDARD FOR THE SCIENTIFIC CONFIDENCE AND LOCATIONAL ACCURACY OF GEOLOGIC MAP FEATURES

In response to reviewers' comments, we have implemented a new standard (concepts and terminology) for the scientific confidence and the locational accuracy of geologic map features (note that, at this time, we have applied these new concepts only to line features). Scientific confidence expresses a geologist's level of certainty about the identity or perhaps even the existence of a feature. Locational accuracy is based on the relation between a feature's location in the field and its position on the base map. For a more detailed discussion of these concepts and their associated terminology, please refer to Section 4 ("Scientific Confidence and Locational Accuracy of Geologic Features") in the accompanying introductory text.

FEATURE ATTRIBUTES FOR SCIENTIFIC CONFIDENCE AND LOCATIONAL ACCURACY

The following is a list of the feature attributes (in italics) that are used to express these concepts:

- Scientific Confidence:
 - Identity — '*certain*' or '*questionable*'
 - Existence — '*certain*' or '*questionable*'
- Locational Accuracy:
 - Locatability — '*observable*,' '*inferred*' (between outcrops or beneath rubble or vegetation), or '*concealed*' (beneath overlying map unit, ice, or water)
 - Zone of Confidence — [*value, in ground units*]; [*unit of measurement*]
 - Positioning Confidence — '*within zone of confidence*' or '*may not be within zone of confidence*'